

Frobenius Minimality, Minimum-Layer Uniqueness, and an Infinite Family of Huneke–Wiegand Counterexamples

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Provenance. Son Pham discovered the counterexample; Professor Craig Huneke independently verified it. This paper contributes certified Frobenius minimality, minimum-layer uniqueness, a separate explicit infinite family, and its uniform endomorphism, type, trace, conductor, stability, and reduction anatomy.

Abstract

Son Pham publicly identified a counterexample to the Huneke–Wiegand conjecture in the class of two-generated monomial ideals over symmetric numerical semigroup rings, subsequently verified by Professor Craig Huneke. We preserve that discovery priority and answer two different questions: how small can such an example be, and does the same semigroup/ideal class contain an explicit infinite family? We prove by proof-carrying exact computation that the least Frobenius number is 181, attained by Pham’s semigroup at ideal shift 14. First, an implementation of the complete Blanco–Rosales tree exhausts all 48,954 symmetric semigroups with Frobenius number below 69 and all 1,503,391 associated gap cases. An independent Boolean route checks 1,156 fixed-pair UNSAT certificates for the same range. We then introduce a one-hot shift-selector CNF that asks one existence question per Frobenius number. Its DRAT certificates exclude every odd value from 69 through 179, while the formula at 181 is satisfiable and decodes exactly to the public semigroup and shift. A separately generated fixed-pair formula returns the same model. An independent audit rehashes all 228 files in the minimality search, totaling 760,081,739 bytes, rechecks both SAT models, and reproduces the strict-order manifest with no mismatch. Complete theorem-tree enumeration at $F = 69, 71, 73, 75$ provides an object-level cross-check on another 79,790 semigroups and 2,933,163 gap cases. A second projected enumeration proves that shift 14 is the only feasible shift at $F = 181$ and that the public membership vector is the only semigroup there. Both terminal DRAT proofs are freshly rechecked in an independent 12-file audit. Thus Pham’s pair is the unique normalized pair attaining the Frobenius minimum in this class. Finally, for every integer $p \geq 4$ we construct a symmetric numerical semigroup Γ_p of multiplicity $24p$, Frobenius number $78p - 1$, conductor $78p$, and embedding dimension $11p$, for which $I_p = (t^{24p}, t^{30p})$ is nonprincipal and rigid. This family theorem is deductive: seven exact interval-sum identities prove closure, symmetry, and $D_s = E_s + E_{\bar{s}}$; a finite sweep through $p = 300$ and an independently implemented audit are supporting checks. For every family member we also

determine the value semigroup of the endomorphism ring exactly. It has multiplicity $24p$, Frobenius number $54p - 1$, conductor $54p$, genus $38p - 1$, and embedding dimension $12p$. It is nonsymmetric, and the endomorphism ring is strictly larger than the original ring. Recent endomorphism-center criteria then show uniformly that the ideal remains rigid over its endomorphism ring but is not reflexive there, with the adjacent Ext and Tor obstruction groups nonzero. We further prove that this endomorphism semigroup has Cohen–Macaulay type and reduced type $10p$, hence maximal reduced type, and that its completed semigroup ring is not almost Gorenstein. Finally, the traces of the ideal and its endomorphism ring both equal the conductor of the finite birational extension. We determine that common value ideal exactly and prove the identity

$$\ell_{R_p}(R_p/(R_p : E_p)) = \ell_{R_p}(E_p/R_p) = p + 1.$$

The equality of the two lengths is a general consequence of one-dimensional Gorenstein local duality, while the exact common ideal and its value $p + 1$ are the family-specific data. We further prove that this conductor is nonstable and compute its complete one-step stability defect:

$$\ell_{R_p}(T_p^2/t^{4s}T_p) = 14p.$$

The full reduction sequence is also exact: $t^{4s}R_p$ is a minimal reduction of T_p with reduction number four, successive Sally-quotient lengths $23p - 1, 14p, 2p, 1, 0$, and Hilbert–Samuel coefficients $e_0(T_p) = 24p$ and $e_1(T_p) = 39p$. The results remain confined to two-generated monomial ideals over symmetric numerical semigroup rings and do not classify arbitrary Gorenstein domains and modules.

1 Introduction

Let R be a one-dimensional Gorenstein local domain and let M be a nonzero finitely generated non-free module. The Huneke–Wiegand conjecture predicted nontrivial torsion in $M \otimes_R \operatorname{Hom}_R(M, R)$ [1]. For a numerical semigroup $\Gamma \subseteq \mathbb{N}$, the semigroup ring $\mathbb{k}[t^\Gamma]$ is Gorenstein exactly when Γ is symmetric. This specialization converts part of the module problem into finite additive combinatorics. García-Sánchez and Leamer proved broad positive classes and reported a NumericalSgps check for every symmetric semigroup with Frobenius number below 69 [2].

In 2026, Son Pham published a symmetric numerical semigroup and a two-generated monomial ideal for which the predicted torsion is absent [5]. Professor Craig Huneke supplied an independent verification archived with the public candidate record. That example has Frobenius number 181. The discovery settles the original conjecture negatively, but does not by itself say whether 181 is small, minimal, or typical.

This paper reports computer-assisted classification theorems and a deductive family theorem. Claims are labeled [MV] when they are verified by exact machine computation and independently checked artifacts, and [D] when derived in the text. No conjectural claim is needed for the main theorem.

Contributions.

1. We give a self-contained finite-window reduction and a direct proof that the selector CNF is equisatisfiable with existence of a rigid two-generated monomial ideal.
2. We reproduce the published $F < 69$ boundary by two independent complete routes: the Blanco–Rosales tree and checked DRAT proofs.

3. We certify UNSAT for every odd $F = 69, 71, \dots, 179$, then recover the exact public model at $(F, s) = (181, 14)$.
4. We independently rehash and inspect the complete search artifact set, and cross-check the first four new odd Frobenius values by explicit theorem-tree enumeration.
5. We classify the complete minimum layer by projected enumeration: the only feasible shift at $F = 181$ is 14, and the public semigroup is the only membership vector at that shift.
6. We identify two structural boundaries: the public semigroup lies outside the generalized arithmetic-sequence positive family, and the visible fixed-offset generator blocks do not extend to a naive parametric family.
7. We use the failed fixed-width ansatz to derive and prove an explicit infinite family for every integer $p \geq 4$, with a separate exact implementation and independent audit.
8. We compute the endomorphism overring of every family member and prove that the nonreflexive Ext/Tor escape identified in the public seed persists uniformly.
9. We compute the complete pseudo-Frobenius set of every endomorphism semigroup, proving unbounded maximal reduced type and a uniform non-almost-Gorenstein boundary.
10. We compute both trace ideals and the conductor exactly, obtaining a common ideal whose colength equals the birational extension length, namely $p + 1$.
11. We separate that general duality balance from the family-specific calculation, prove that the common conductor is nonstable, and compute its exact stability defect $14p$.
12. We determine every later power of the conductor, proving reduction number four, the full Sally-quotient profile, and Hilbert–Samuel coefficients $(e_0, e_1) = (24p, 39p)$.

2 The finite semigroup problem

Definition 2.1. A *numerical semigroup* is a cofinite additive submonoid $\Gamma \subseteq \mathbb{N}$. Its Frobenius number is $F(\Gamma) = \max(\mathbb{N} \setminus \Gamma)$. It is *symmetric* when

$$n \in \Gamma \iff F(\Gamma) - n \notin \Gamma \quad (0 \leq n \leq F(\Gamma)).$$

Fix a symmetric semigroup Γ with Frobenius number F and a positive gap $s \notin \Gamma$. Normalize the corresponding two-generated relative ideal to $(0, s)$. Define

$$\begin{aligned} E_s &= \{n \in \mathbb{N} : n, n + s \in \Gamma\}, \\ D_s &= \{n \in \mathbb{N} : n, n + s, n + 2s \in \Gamma\}. \end{aligned}$$

The García–Sánchez–Leamer reduction identifies the counterexample condition with

$$D_s = E_s + E_s. \tag{1}$$

Equivalently, every length-two arithmetic sequence of step s in Γ decomposes as a sum of two length-one sequences [2, 4].

Lemma 2.2 (Automatic inclusion, [D]). *For every numerical semigroup Γ and every positive integer s , $E_s + E_s \subseteq D_s$.*

Proof. If $a, b \in E_s$, then $a, a + s, b, b + s \in \Gamma$. Closure gives $a + b \in \Gamma$, $a + b + s = a + (b + s) \in \Gamma$, and $a + b + 2s = (a + s) + (b + s) \in \Gamma$. Hence $a + b \in D_s$. $\square$

Lemma 2.3 (Finite window, [D]). *Equation (1) holds if and only if $D_s \cap [0, 2F + 1] \subseteq E_s + E_s$.*

Proof. Only the reverse inclusion needs checking by Lemma 2.2. Every integer at least $F + 1$ belongs to Γ , hence to E_s and D_s . Let $m = \min E_s$; then $m \leq F + 1$. For every $n \geq m + F + 1$, both m and $n - m$ lie in E_s , so $n \in E_s + E_s$. Since $m + F + 1 \leq 2F + 2$, every $n \geq 2F + 2$ is automatic. The displayed finite window is therefore complete. $\square$

Lemma 2.4 (Parity, [D]). *The Frobenius number of a symmetric numerical semigroup is odd.*

Proof. If F were even, symmetry at $n = F/2$ would require $F/2$ to belong to Γ exactly when it does not belong to Γ . $\square$

3 The public model and independent calibration

Pham's public semigroup is

$$\Gamma = \langle 56, 57, 58, 63, 64, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, \\ 87, 89, 90, 93, 95, 96, 97 \rangle,$$

with the ideal generated by t^{56} and t^{70} , hence normalized shift $s = 14$ [5]. An independent Singular/4ti2 calculation and a separate standard-library finite route verify $F = 181$, conductor 182, genus 91, symmetry, and equality (1) [MV]. The Singular route uses a 322-generator toric standard basis; both reduced ideal differences vanish. The hypersurface control $\langle 4, 5 \rangle$ rejects equality with explicit residues [MV].

The endomorphism overring of the ideal has value semigroup

$$v(\text{End}_R(I)) = \Gamma \cup \{101, 107, 181\},$$

with Frobenius number 125, genus 88, and type 24 [MV]. This explains why positive criteria that require a Gorenstein endomorphism ring do not apply, but it does not imply minimality.

4 Complete reproduction below 69

For fixed odd F , irreducible numerical semigroups are exactly the symmetric ones. Blanco and Rosales construct all of them as a rooted tree [3]. Starting from

$$C(F) = \{0, (F + 1)/2, (F + 1)/2 + 1, \dots\} \setminus \{F\},$$

a child replaces a suitable minimal generator $x > F/2$ by $F - x$. Their theorem supplies exact child conditions, completeness, and a unique parent.

Our direct implementation validates every node for closure, symmetry, genus, minimal generators, and its theorem parent. It then checks every gap by Lemma 2.3 and retains a first element of $D_s \setminus (E_s + E_s)$ whenever equality fails. The full range $F = 1, 3, \dots, 67$ contains 48,954 semigroups

and 1,503,391 gap cases; none is rigid [MV]. At $F = 11$ the implementation returns the six nodes listed by Blanco–Rosales, while an invalid mutation is rejected by closure at $(1, 1)$.

An independent route generates one fixed-pair DIMACS formula for each (F, s) , including shifts that are forced not to be gaps. CaDiCaL emits a DRAT proof for every UNSAT result, and DRAT-trim checks each proof against the exact CNF [12, 13]. All 1,156 formulas are UNSAT and independently verified [MV]. A committed auditor rehashes 4,624 external files totaling 211,671,241 bytes with zero mismatch.

5 One selector formula per Frobenius value

The fixed-pair route duplicates the same semigroup constraints once per shift. We instead introduce Boolean membership variables h_n for $0 \leq n \leq F$ and one-hot variables q_s for $1 \leq s \leq F$.

5.1 Semigroup and selector clauses

The CNF contains units h_0 and $\neg h_F$, the symmetry clauses

$$(h_n \vee h_{F-n}) \wedge (\neg h_n \vee \neg h_{F-n}),$$

and closure clauses $\neg h_a \vee \neg h_b \vee h_{a+b}$ for $a + b \leq F$. The selectors satisfy one at-least-one clause, all pairwise at-most-one clauses, and $q_s \Rightarrow \neg h_s$. Thus every model chooses exactly one gap.

For $0 \leq n \leq 2F + 1$, guarded Tseitin clauses impose, under the selected q_s ,

$$\begin{aligned} e_n &\longleftrightarrow h_n \wedge h_{n+s}, \\ d_n &\longleftrightarrow h_n \wedge h_{n+s} \wedge h_{n+2s}, \end{aligned}$$

where h_j is the constant true for $j > F$. For each unordered decomposition $n = a + b$, a variable $p_{n,a,b}$ is equivalent to $e_a \wedge e_b$. Finally,

$$d_n \Rightarrow \bigvee_{a+b=n} p_{n,a,b}.$$

The automatic inclusion from Lemma 2.2 need not be duplicated in the formula.

Proposition 5.1 (Selector equisatisfiability, [D]). *For positive odd F , the selector CNF is satisfiable if and only if there exist a symmetric numerical semigroup Γ with Frobenius number F and a gap s satisfying $D_s = E_s + E_s$.*

Proof. The semigroup clauses decode exactly a symmetric semigroup of Frobenius number F , and the one-hot clauses decode a unique gap. Because exactly one guard is true, all e_n and d_n equal the definitions for that gap. The p equivalences and final disjunctions impose $D_s \subseteq E_s + E_s$ through $2F + 1$. Lemmas 2.2 and 2.3 give global equality. Conversely, from a rigid pair assign h and q from the pair, assign e, d by their definitions, and assign each p by its conjunction. Every clause is then true. $\square$

The selector encoding passes two adversarial calibrations [MV]. Every odd $F \leq 67$ is UNSAT with an accepted proof, while $F = 181$ is SAT. The latter formula has 34,397 variables and 363,912 clauses, selects $s = 14$, and decodes exactly to the public membership vector. A newly generated fixed-pair formula at $(181, 14)$ returns the same vector.

6 The certified search and minimality theorem

We scan odd F in strict increasing order. Table 1 summarizes the closed run.

Table 1: Selector-CNF minimality search.

Quantity	Value
Odd Frobenius values requested	57 (69 through 181)
UNSAT with accepted DRAT proof	56 (69 through 179)
Validated SAT models	1 (181, shift 14)
UNKNOWN or timeout	0
Wall time	4,303.86 seconds
CNF bytes	169,745,433
DRAT proof bytes	586,785,257
Slowest solve	$F = 175$, 218.83 seconds

Theorem 6.1 (Frobenius minimality, [MV+D]). *Among symmetric numerical semigroups admitting a nonprincipal two-generated monomial ideal that is a counterexample to the Huneke–Wiegand prediction, the least Frobenius number is 181. It is attained by Pham’s public example at shift 14.*

Proof. By Lemma 2.4, only odd Frobenius values occur. Complete tree enumeration and the independent fixed-pair proof suite exclude every odd $F \leq 67$. Proposition 5.1 and the accepted selector DRAT proofs exclude every odd F from 69 through 179. The validated selector and fixed-pair models at (181, 14) provide existence. These cases exhaust all smaller positive Frobenius values. $\square$

The search audit independently rehashes all 228 expected CNF, proof, solver-log, and checker-log files, totaling 760,081,739 bytes. It also reconstructs both SAT membership masks, validates closure, symmetry, and rigidity, checks strict odd ordering, and recomputes aggregate SHA-256

0f580de2707a00fdd52e1b3c04e7767b97ce7b0a826593b119e9a49ae04da743.

No file is missing, no hash differs, and no status marker or semantic check fails [MV].

7 Complete classification at the minimum

The selector search proves existence at $F = 181$ but, by itself, does not classify every model at that value. We close this gap in two projected layers. First, after discovering a valid selector model at shift s , we add the unit clause $\neg q_s$. Second, for each feasible shift we enumerate the fixed-shift formula and block each complete membership projection.

Lemma 7.1 (Projected blocker, [D]). *Let $x_1, \dots, x_k$ be distinct Boolean variables with complete assignment a . The clause*

$$B(a) = \bigvee_{a_i=1} \neg x_i \vee \bigvee_{a_i=0} x_i$$

is false exactly on assignment a . Its truth is independent of every variable outside the projection.

Proof. Every displayed literal is false under a . Any other complete assignment differs at some coordinate j , where the corresponding displayed literal is true. Variables absent from the clause cannot affect either statement. $\square$

For semigroup enumeration, $B(a)$ contains one signed literal for every $h_0, \dots, h_{181}$. Consequently, it excludes one mathematical membership vector and every auxiliary extension of that vector, while preserving every different semigroup. Blocking only the positive coordinates would not have this property.

Theorem 7.2 (Minimum-layer uniqueness, [MV+D]). *Up to translation and interchange of the two monomial generators, Pham’s pair is the unique nonprincipal two-generated monomial-ideal counterexample over a symmetric numerical semigroup ring with Frobenius number 181.*

Proof. The unblocked selector formula returns the exact public pair at shift 14. After adding $\neg q_{14}$, the formula is UNSAT and its DRAT proof is accepted, so no other shift is feasible. The fixed (181, 14) formula returns the exact public membership vector. After adding its complete 182-literal blocker from Lemma 7.1, that formula is UNSAT and its DRAT proof is accepted. Thus no second semigroup exists at the only feasible shift. Translation and interchange reduce any two monomial generators to one positive normalized shift, which proves the statement. $\square$

Table 2: Certificate summary for the complete $F = 181$ layer.

Quantity	Value
Feasible shifts	1, namely {14}
Normalized rigid pairs	1
Distinct membership vectors	1
Support terminal proof bytes	45,867,741
Fixed-class terminal proof bytes	1,608,691
Audited external files	12
Audited external bytes	63,609,504
Fresh proof-recheck time	232.88 seconds

The independent auditor reconstructs all four CNFs byte-for-byte, decodes both SAT logs, reruns closure, symmetry, minimal-generator and full-window/tail rigidity checks, and freshly checks both terminal proofs [MV]. It accounts for every external file and reproduces manifest SHA-256

02d1265b75e886bc6ec693e60b367b661476170c755e38a220342396564ef5d3.

The compact classification aggregate is

b37d9410fa07c00f22ccc0f83796644db3b44f0dd654c06933aa5d22cb1ed788.

8 Independent object-level checks beyond the old frontier

The theorem-tree route was continued beyond 67 without consulting SAT models. Table 3 lists the first four new odd values.

This is not needed to trust a DRAT derivation, but it attacks the custom encoding at the object level and supplies explicit failed-rigidity witnesses for every tested gap. The routes agree.

Table 3: Complete Blanco–Rosales tree cross-checks beyond the published boundary.

F	Symmetric semigroups	Gap cases	Rigid cases	Seconds
69	11,276	394,660	0	137.00
71	19,812	713,232	0	235.37
73	25,405	939,985	0	333.21
75	23,297	885,286	0	330.10
Total	79,790	2,933,163	0	1,035.68

9 From a failed ansatz to an infinite family

Landeros et al. prove the positive result for semigroups generated by a generalized arithmetic sequence $\langle a, ah + d, \dots, ah + kd \rangle$ [4]. The public semigroup cannot lie in this class [D]. Its multiplicity forces $a = 56$. Since $57 \in \Gamma$ and no sum of two positive semigroup elements is below 112, the presentation forces $h = d = 1$. Since $63 \in \Gamma$, it would then include 59, 60, 61, 62, but all four are gaps.

The seed also displays $56 = 4s$ and $181 = 13s - 1$ at $s = 14$, with generator blocks

$$4s + \{0, 1, 2, 7, 8\}, \quad 5s + [0, s - 1], \quad 6s + \{3, 5, 6, 9, 11, 12, 13\}.$$

A predeclared exact sweep refutes the naive fixed-offset lift [MV]. Only $s = 14$ passes among even $s \leq 100$. At $s = 16$, the generated semigroup has $F = 205$ rather than 207; at $s = 28$ the predicted Frobenius value returns but symmetry fails. This negative result prevents the minimality theorem from being overinterpreted as evidence of an immediate family.

We next retained only the structural scaffold $m = 4s$, $F = 13s - 1$, the full interval $[5s, 6s - 1] \subseteq \Gamma$, and rigidity at shift s . A proof-carrying Boolean search certifies that the scaffold is infeasible at $s = 16, 18$ and returns independently validated models for every even $s = 20, 22, \dots, 40$ [MV]. The eleven non-seed models justified a separate extraction stage, but did not themselves prove recurrence. The first fixed-width interval formula passed three parameters and then failed: at $s = 38$, $9s + 7$ lies in D_s but not in $E_s + E_s$. For every later parameter in that formula, carried sums from the first block end at residue 6, while the next summand block begins at residue at least 8. This uniform obstruction suggests widening the first interval with the parameter.

9.1 The parametric blocks

Fix an integer $p \geq 4$ and put $s = 6p$. All intervals below are integer intervals. Define subsets of $[0, s - 1]$ by

$$\begin{aligned} A_p &= [0, p] \cup [3p, 4p - 2], \\ B_p &= ([p + 1, 3p - 1] \setminus \{2p - 1\}) \cup \{4p\} \cup [5p - 1, 6p - 1], \\ C_p &= [0, 2p] \cup [3p, 5p - 2]. \end{aligned}$$

Define $\Gamma_p \subseteq \mathbb{N}$ by the membership blocks

$$\{0\}, \quad 4s + A_p, \quad [5s, 6s - 1], \quad 6s + B_p, \quad 8s + C_p, \quad [9s, 13s - 2], \quad [13s, \infty), \quad (2)$$

with every unlisted nonnegative integer below $13s$ a gap.

For subsets of $[0, s - 1]$, let $\text{low}(X + Y)$ denote the sums below s , and let $\text{carry}(X + Y)$ denote the sums at least s after subtraction of s .

Lemma 9.1 (Residue identities, [D]). *For every $p \geq 4$,*

$$\begin{aligned} \text{low}(A_p + A_p) &= C_p, & \text{carry}(A_p + A_p) &= [0, 2p - 4], \\ \text{low}(A_p + B_p) &= [p + 1, 6p - 1], & \text{carry}(A_p + B_p) &= [0, 4p - 3], \\ \text{low}(B_p + B_p) &= [2p + 2, 6p - 2], & \text{carry}(B_p + B_p) &= [0, 6p - 2], \\ \text{low}(A_p + C_p) &= [0, 6p - 2]. \end{aligned}$$

Moreover,

$$[0, 2p - 4] \cup [p + 1, 6p - 1] = [0, 6p - 1], \quad (3)$$

$$[0, 4p - 3] \cup [2p + 2, 6p - 2] = [0, 6p - 2]. \quad (4)$$

Proof. Split A_p into its two displayed intervals and split the punctured component of B_p at $2p - 1$. Pairwise interval addition, followed by separating sums below and above $s = 6p$, gives the seven displayed identities. The singleton $4p$ fills the endpoint removed from the middle of B_p . All interval joins are immediate except (3): its two pieces meet exactly when $2p - 3 \geq p + 1$, equivalently $p \geq 4$. For (4), the overlap condition is $4p - 2 \geq 2p + 2$, which also holds for $p \geq 4$. $\square$

Theorem 9.2 (Infinite family, [D]). *For every integer $p \geq 4$, the set Γ_p in (2) is a symmetric numerical semigroup with multiplicity $24p$, Frobenius number $78p - 1$, conductor $78p$, and embedding dimension $11p$. Let*

$$R_p = \mathbb{k}[t^{\Gamma_p}]_{(t^n : n \in \Gamma_p \setminus \{0\})}, \quad I_p = (t^{24p}, t^{30p})R_p.$$

Then R_p is Gorenstein and I_p is nonprincipal and rigid. Hence the pairs (R_p, I_p) form an infinite family of counterexamples in the two-generated monomial-ideal class.

Proof. We first prove the semigroup assertions. The only sums of positive displayed members that can stay below the conductor have starting levels among 4, 5, 6, 8. Lemma 9.1 handles the non-full target blocks: $A_p + A_p$ produces $8s + C_p$ and part of level 9; $A_p + B_p$, $B_p + B_p$, and $A_p + C_p$ land exactly inside the declared high blocks and never produce the Frobenius residue $s - 1$ in level 12. Sums involving the full level 5 block either fill a declared level or enter the conductor tail. Thus Γ_p is closed.

Conversely, the lower three blocks generate every displayed member. The identity $\text{low}(A_p + A_p) = C_p$ generates level 8. The presence of $0 \in A_p$ makes $(4s + A_p) + [5s, 6s - 1]$ fill level 9, and two level-5 elements fill level 10 and residues $0, \dots, s - 2$ of level 11; the residue $s - 1 \in B_p$ fills the remaining point. Three elements from $4s + A_p$ generate residues $0, \dots, 3p$ of level 12, while $\text{low}(B_p + B_p) = [2p + 2, s - 2]$ completes that level through $13s - 2$. Adding $4s$ to $[9s, 13s - 2]$ gives $[13s, 17s - 2]$, and $(4s + 1) + (13s - 2) = 17s - 1$. These are $4s$ consecutive generated integers, so repeated addition of the multiplicity gives the entire tail.

Therefore the multiplicity is $4s = 24p$, the conductor is $13s = 78p$, and the Frobenius number is $13s - 1 = 78p - 1$. Every member of the three lower blocks lies below $7s$, whereas a sum of two positive members is at least $8s$; all are minimal generators. Since $|A_p| = 2p$, $|B_p| = 3p$, and the full block has size $6p$, the embedding dimension is $11p$.

Symmetry is blockwise about $F = 13s - 1$. The residue-reflection complement of A_p is exactly C_p . The full level-5 block reflects to the empty level-7 block. In B_p , exactly one residue from each pair $\{r, s - 1 - r\}$ occurs: the puncture $2p - 1$ pairs with the singleton $4p$, and the remaining displayed intervals pair with their omitted intervals. The gaps below $4s$ reflect to the full range $[9s, 13s - 2]$, and 0 reflects to the gap F . Hence Γ_p is symmetric, so R_p is Gorenstein.

It remains to prove rigidity. Normalize I_p to $J_p = (1, t^s)$ and use E_s, D_s from Section 2. Direct inspection of (2) gives the nontrivial residue layers

$$\begin{aligned} E_4 &= A_p, & E_5 &= B_p, & E_8 &= C_p, \\ E_9 &= E_{10} = [0, s-1], & E_{11} &= E_{12} = [0, s-2], \\ D_8 &= C_p, & D_9 &= [0, s-1], \\ D_{10} &= D_{11} = D_{12} = [0, s-2]. \end{aligned}$$

All lower D layers are empty; in particular D_4 is empty because $A_p \cap B_p = \emptyset$. Lemma 9.1 and (3)–(4) compute the five load-bearing layers of $E_s + E_s$:

$$\begin{aligned} D_8 &= \text{low}(A_p + A_p), & D_9 &= \text{carry}(A_p + A_p) \cup \text{low}(A_p + B_p), \\ D_{10} &= \text{carry}(A_p + B_p) \cup \text{low}(B_p + B_p), & D_{11} &= \text{carry}(B_p + B_p), & D_{12} &= \text{low}(A_p + C_p). \end{aligned}$$

At levels 13 and 14, the zero residue in A_p translates full high E blocks. At levels 15 and 16, those high blocks omit only $s-1$, and the residues $0, 1 \in A_p$ fill both the interval and its final endpoint. Level 17 is filled by $E_4 + E_{13}$; levels 18, 19, and 20 by $E_9 + E_9$, $E_9 + E_{10}$, and $E_{10} + E_{10}$; and level 21 by $E_8 + E_{13}$ because $0 \in C_p$. Every level from 22 onward is filled by $E_9 + E_q$ with $q \geq 13$, where the conductor makes E_q full. Thus $D_s \subseteq E_s + E_s$; Lemma 2.2 gives equality.

Finally, s is a gap. If J_p were principal, write $J_p = xR_p$. Since $1 \in J_p$, there is $r \in R_p$ with $xr = 1$, so $rJ_p = R_p$ and $(r, rt^s) = R_p$. In the local ring, one of these two generators must be a unit. Positive valuation excludes rt^s , so r is a unit and $t^s \in R_p$, a contradiction. The ideal is therefore nonprincipal and rigid. The multiplicities $24p$ are distinct, so the family contains infinitely many distinct rings. $\square$

9.2 Exact support and boundaries

The proof of Theorem 9.2 is independent of a finite sweep. As adversarial support, the implementation evaluated every $p = 2, 3, \dots, 300$ [MV]. Exactly the predicted boundary values $p = 2, 3$ fail, while all 297 values $p \geq 4$ pass the seven identities, closure, generation, symmetry, invariants, and full-window/tail rigidity. The instances $p = 4, 5$ reproduce independently generated Boolean models byte-for-byte. Two corrupted positive controls are rejected. A separate auditor reconstructs all 297 positive membership hashes from the formula and performs fresh semantic checks at $p = 4, 5, 17, 73, 151, 300$. The campaign aggregate is

81d5a8eb6cf2e848807323e3b0bdba58c464779d25cdc788cef027585540dce2.

The independent-audit aggregate is

eb2aaf17650ed99f4e220a43c53bdd8835c82688a37567bb154c30a1ae520ce9.

The family remains outside the generalized-arithmetic-sequence positive class [D]. Membership of $4s+1$ forces unit step, while membership of $4s+3p$ would then force the actual gap $4s+p+1$. Theorem 9.2 does not say that every model in the Boolean scaffold belongs to this family, or that $p=4$ is a global threshold outside the displayed formula.

10 Uniform endomorphism overrings

Normalize I_p to $J_p = (1, t^s)$ and write

$$V_p = v(J_p) = \Gamma_p \cup (s + \Gamma_p), \quad \Lambda_p = v(\text{End}_{R_p}(J_p)).$$

Define

$$Q_p = [p + 1, 2p - 2] \cup \{2p, 4p\}.$$

Theorem 10.1 (Uniform endomorphism anatomy, [D]). *For every integer $p \geq 4$,*

$$\Lambda_p = \Gamma_p \cup (7s + Q_p) \cup \{13s - 1\}.$$

This numerical semigroup has multiplicity $24p$, Frobenius number $54p - 1$, conductor $54p$, genus $38p - 1$, and embedding dimension $12p$. It is nonsymmetric, so the localized semigroup ring $E_p = \text{End}_{R_p}(J_p)$ is not Gorenstein. Moreover,

$$\begin{aligned} \text{Ext}_{E_p}^1(J_p, J_p) &= 0, & J_p &\text{ is not reflexive over } E_p, \\ \text{Ext}_{E_p}^1(J_p, E_p) &\neq 0, & \text{Ext}_{E_p}^2(J_p, J_p) &\neq 0, \\ \text{Tor}_1^{R_p}(J_p, E_p) &\neq 0. \end{aligned}$$

Proof. Let G_k and V_k denote the residue sets at levels ks in Γ_p and V_p , respectively. From (2),

$$V_0 = V_1 = \{0\}, \quad V_2 = V_3 = \emptyset, \quad V_4 = A_p, \quad V_5 = V_6 = [0, s - 1], \quad V_7 = B_p, \quad V_8 = C_p,$$

and $V_k = [0, s - 1]$ for every $k \geq 9$. An exponent $n = ks + r$ stabilizes J_p exactly when both n and $n + s$ belong to V_p . Thus the level- k residue set of Λ_p is $V_k \cap V_{k+1}$.

All adjacent intersections reproduce the corresponding Γ_p block except at level 7 and the old Frobenius point. Directly,

$$B_p \cap C_p = ([p + 1, 2p] \setminus \{2p - 1\}) \cup \{4p\} = Q_p.$$

Every level from 9 onward is full, so the only other added value is $13s - 1$. This proves the formula for Λ_p .

The first positive value remains $4s = 24p$. Level 8 still omits residue $s - 1$, while level 9 and all later levels are full. Hence the Frobenius number is $9s - 1 = 54p - 1$ and the conductor is $9s = 54p$. Counting gaps by levels gives

$$(4s - 1) + (s - |A_p|) + (s - |B_p|) + (s - |Q_p|) + (s - |C_p|) = 38p - 1.$$

A symmetric semigroup with this Frobenius number would have genus $27p$, so Λ_p is nonsymmetric.

The $11p$ generators of Γ_p and the p new values $7s + Q_p$ all lie below twice the multiplicity, hence are minimal. For any $q \in Q_p$, the level-5 block contains $5s + (s - 1 - q)$ and

$$(7s + q) + (5s + (s - 1 - q)) = 13s - 1.$$

Thus the old Frobenius point is generated and there are exactly $12p$ minimal generators.

Finally, R_p is a one-dimensional Gorenstein local domain and J_p is faithful, torsion-free, two-generated, nonprincipal, and rigid by Theorem 9.2. The ring E_p is commutative and is strictly larger than R_p because Q_p is nonempty. Proposition 4.1(2) of Dey and Lyle makes rigidity descend to E_p ; the contrapositives of their Theorems 4.2–4.4 give the displayed nonreflexivity, Ext, and Tor conclusions [6]. $\square$

The symbolic argument is independent of a parameter sweep. As adversarial support, two exact implementations agree for every $p = 4, \dots, 300$ [MV]. A separate auditor rehashes all 297 rows, reconstructs complete semantic windows at $p = 4, 5, 17, 73, 151, 300$, and rejects a missing Q_p value and a missing terminal singleton. The deterministic campaign and audit aggregates are

e21926a689178a6c70b3b6e8319053edd0fd13f164ced9565d3b976e6159c0b0

and

2ed711045ad83a3b47fb3e71d4c75ae9bfa9be1a5dd9a8c4072d5f170510343b.

11 Pseudo-Frobenius anatomy and reduced type

For a numerical semigroup H , write

$$\text{PF}(H) = \{f \notin H : f + h \in H \text{ for every } h \in H \setminus \{0\}\}.$$

Its Cohen–Macaulay type is $|\text{PF}(H)|$. For complements inside $[0, s - 1]$, define

$$\begin{aligned} B_p^c &= [0, p] \cup \{2p - 1\} \cup [3p, 4p - 1] \cup [4p + 1, 5p - 2], \\ Q_p^c &= [0, p] \cup \{2p - 1\} \cup [2p + 1, 4p - 1] \cup [4p + 1, 6p - 1], \\ C_p^c &= [2p + 1, 3p - 1] \cup [5p - 1, 6p - 1]. \end{aligned}$$

Theorem 11.1 (Type anatomy, [D]). *For every integer $p \geq 4$,*

$$\text{PF}(\Lambda_p) = (6s + B_p^c) \cup (7s + Q_p^c) \cup (8s + C_p^c).$$

Consequently the type and reduced type of $\mathbb{k}[[\Lambda_p]]$ are both $10p$. Thus the ring has maximal reduced type. The semigroup is not almost symmetric, and $\mathbb{k}[[\Lambda_p]]$ is not almost Gorenstein.

Proof. The multiplicity of Λ_p is $4s$ and its conductor is $9s$. Every gap in the final multiplicity window $[5s, 9s - 1]$ is therefore pseudo-Frobenius. Since level 5 is full, these gaps are precisely the complements of B_p, Q_p, C_p at levels 6, 7, and 8. Their cardinalities are $3p, 5p, 2p$.

It remains to exclude a pseudo-Frobenius number below $5s$. The full level-5 generator block gives explicit witnesses. For a gap r at level 0 with $r > 0$, add $5s + (s - r)$ to reach the level-6 gap of residue zero. At level 1, use $5s$ when $r = 0$ and $5s + (s - r)$ otherwise. At levels 2 and 3, add $5s + (s - 1 - r)$ to reach residue $s - 1$ in a missing later block.

For a level-4 gap $4s + r$, if $r \notin C_p$ add the generator $4s$. The remaining residues lie in

$$C_p \setminus A_p = [p + 1, 2p] \cup [4p - 1, 5p - 2].$$

In the first interval add $4s + (2p + 1 - r)$; in the second add $4s + (5p - 1 - r)$. Both added residues lie in $[1, p] \subset A_p$, while the resulting level-8 residue is a gap of C_p . Thus no lower gap is pseudo-Frobenius, proving the formula.

Reduced type counts the gaps in the last multiplicity window, so it is also $10p$ [7]. Finally, using $g(\Lambda_p) = 38p - 1$ and $F(\Lambda_p) = 54p - 1$,

$$2g - (F + \text{type}) = 12p - 1 > 0.$$

The almost-symmetric genus identity therefore fails, giving the final claims. $\square$

Two exact implementations confirm the displayed set for all $p = 4, \dots, 300$ [MV]. One checks every gap against the full minimal-generator set; the other extracts maximal elements of the Apéry poset modulo $4s$. The campaign and independent-audit aggregates are

9bed38fb1c786c3740e000dde7ea7d79a7e7c83fa584ff12fc2c4623b5d503ec

and

0315c4c22c41e0d2b8a5abb27f717a4d4a6f7356ef30ca388206d739e0de2c37.

12 Exact trace and conductor ideals

Let $E_p = \text{End}_{R_p}(J_p) = \mathbb{k}[[\Lambda_p]]$ and define the reflected block

$$H_p = (s - 1) - Q_p = \{2p - 1, 4p - 1\} \cup [4p + 1, 5p - 2].$$

Theorem 12.1 (Exact trace and conductor, [D]). *For every integer $p \geq 4$,*

$$\text{tr}_{R_p}(J_p) = R_p : E_p = \text{tr}_{R_p}(E_p) = T_p,$$

where

$$\begin{aligned} v(T_p) &= (4s + A_p) \cup (5s + (A_p \cup B_p)) \cup (6s + B_p) \cup (8s + C_p) \\ &\quad \cup [9s, 13s - 2] \cup [13s, \infty). \end{aligned}$$

Moreover,

$$\ell_{R_p}(R_p/T_p) = \ell_{R_p}(E_p/R_p) = p + 1.$$

Proof. Because $1 \in J_p$,

$$v(R_p : J_p) = W_p = \{n \in \Gamma_p : n + s \in \Gamma_p\}.$$

Intersecting adjacent blocks of (2) gives

$$(W_4, W_5, W_6, W_7, W_8) = (A_p, B_p, \emptyset, \emptyset, C_p),$$

with full blocks at levels 9 and 10, blocks $[0, s - 2]$ at levels 11 and 12, and full blocks from level 13 onward. Since W_p is a Γ_p -ideal,

$$v(\text{tr}_{R_p}(J_p)) = W_p \cup (s + W_p),$$

which is exactly the displayed set $v(T_p)$.

For the conductor $R_p : E_p$, only the values in

$$\Lambda_p \setminus \Gamma_p = (7s + Q_p) \cup \{13s - 1\}$$

add constraints. The terminal singleton excludes $n = 0$ and imposes no condition on positive $n \in \Gamma_p$. Every value at level 4 passes the $7s + Q_p$ translates: a failure in the sole level-12 gap would require $a + q = 2s - 1$, but $a + q \leq 8p - 2 < 12p - 1$. At level 5, residue r fails exactly when $r + q = s - 1$ for some $q \in Q_p$, hence exactly when $r \in H_p$. All values from level 6 onward translate into the conductor tail. Since

$$[0, s - 1] \setminus H_p = A_p \cup B_p,$$

this proves $v(R_p : E_p) = v(T_p)$. The conductor is an E_p -ideal, so $\text{tr}_{R_p}(E_p) = E_p(R_p : E_p) = R_p : E_p$.

Relative to Γ_p , the common ideal omits exactly $\{0\} \cup (5s + H_p)$, giving colength $p + 1$. The endomorphism formula gives exactly $p + 1$ values in $\Lambda_p \setminus \Gamma_p$, proving the balanced identity. $\square$

The equality of the two traces is consistent with the reflexive-trace criterion of Lindo–Maitra–Zhang [8]; over the present Gorenstein base it is not by itself discriminatory. The equality of the two lengths is also general structure, not a new family-specific mechanism. Indeed, applying $\text{Hom}_{R_p}(-, R_p)$ to $0 \rightarrow R_p \rightarrow E_p \rightarrow E_p/R_p \rightarrow 0$ and using local duality gives

$$\ell_{R_p}(R_p/(R_p : E_p)) = \ell_{R_p}(E_p/R_p),$$

as in Herzog–Kumashiro, Proposition 3.1, Claim 1 [9]. The new family-specific data in Theorem 12.1 are the exact common ideal and the evaluation of both lengths as $p + 1$.

The first declared shorthand used the tail $[9s, \infty)$ and was refuted at $13s - 1$ by the $p = 4$ smoke check [MV]. That point is the Frobenius gap of Γ_p , so no R_p -ideal can contain it. The corrected formula above was then checked by two exact routes for all $p = 4, \dots, 300$, with independent reconstruction at six parameters and three rejected corruptions. The campaign and audit aggregates are

77448398a26b958c66818b7ac4aaa4b542bad11cdba5ec7c8f7fe76db37526e2

and

d55ed876d7918cb4c46d8e5f3894a508d172693bde4b4c76cb67c42fcfbf0ac1.

13 Exact conductor-stability defect

Dey proves that for a finite birational extension $R \subseteq S$ of a one-dimensional Gorenstein local ring, the conductor $R : S$ is stable if and only if S is Gorenstein [10, Corollary 3.7]. Theorem 11.1 gives $\text{type}(E_p) = 10p > 1$, so this criterion already shows that $T_p = R_p : E_p$ is nonstable. The following computation quantifies that failure.

Theorem 13.1 (Exact stability defect, [D]). *For every integer $p \geq 4$,*

$$v(T_p^2) = (8s + C_p) \cup [9s, 13s - 2] \cup [13s, \infty).$$

Moreover,

$$\begin{aligned} v(T_p^2) \setminus v(t^{4s}T_p) = & 8s + ([p + 1, 2p] \cup [4p - 1, 5p - 2]) \\ & \cup 9s + (\{2p - 1, 4p - 1\} \cup [4p + 1, 5p - 2]) \\ & \cup 10s + ([0, p] \cup \{2p - 1\} \cup [3p, 4p - 1] \cup [4p + 1, 5p - 2]) \\ & \cup 11s + [0, s - 1] \\ & \cup 12s + ([2p + 1, 3p - 1] \cup [5p - 1, s - 2]) \\ & \cup \{17s - 1\}. \end{aligned}$$

Consequently,

$$\ell_{R_p}(T_p^2/t^{4s}T_p) = 14p.$$

Proof. Put $D = [0, s - 1]$, $D^- = [0, s - 2]$, and $U_p = A_p \cup B_p$. For residue sets $X, Y \subseteq D$, write $L(X, Y)$ for sums below s and $H(X, Y)$ for sums at least s , translated by $-s$. Expanding the

defining sets into integer intervals and using $[a, b] + [c, d] = [a + c, b + d]$ gives

$$\begin{aligned} L(A_p, A_p) &= C_p, \\ H(A_p, A_p) \cup L(A_p, U_p) &= D, \\ H(A_p, U_p) \cup L(A_p, B_p) \cup L(U_p, U_p) &= D, \\ H(A_p, B_p) \cup H(U_p, U_p) \cup L(U_p, B_p) &= D, \\ H(U_p, B_p) \cup L(B_p, B_p) \cup L(A_p, C_p) &= D^-. \end{aligned}$$

These five identities list all contributions, with carries, to levels 8 through 12 of T_p^2 . They give $(8s + C_p) \cup [9s, 13s - 2]$ and exclude the inherited gap $13s - 1$. Multiplication by the least value $4s$ covers $[13s, 17s - 2] \cup [17s, \infty)$, while

$$17s - 1 = (4s + 1) + (13s - 2)$$

fills the intervening endpoint. This proves the square formula.

Shifting Theorem 12.1 by $4s$ and subtracting its blocks from the square gives the displayed defect. The six block cardinalities are $2p, p, 3p, 6p, 2p - 1, 1$, whose sum is $14p$. $\square$

The first EXP-015 prediction incorrectly replaced the two tail intervals by $[9s, \infty)$. The $p = 4$ smoke gate refuted it at $13s - 1$ before a campaign artifact was written [MV]. EXP-016 retained that gap, proved the corrected formula above, and checked two exact routes for all $p = 4, \dots, 300$. Its campaign and independent-audit aggregates are

8c280bf1db017f678896d473461ae245b4fce57d680be94094b0884556d91853

and

ce4ded03761eb44cba766eafb46926c8ca83d0f7f49a26697afed5ed49905653.

14 Conductor reduction number and Hilbert data

For an ideal I , a subideal Q is a *reduction* when $I^{r+1} = QI^r$ for some r ; the least such r is the reduction number with respect to Q [11].

Theorem 14.1 (Exact conductor reduction sequence, [D]). *For every integer $p \geq 4$, put $Q_p = t^{4s}R_p \subset T_p$. Then Q_p is a minimal reduction of T_p with reduction number four. More precisely,*

n	0	1	2	3	$n \geq 4$
$\ell_{R_p}(T_p^{n+1}/Q_p T_p^n)$	$23p - 1$	$14p$	$2p$	1	0.

The Hilbert–Samuel function satisfies

$$\ell_{R_p}(R_p/T_p^n) = 24pn - 39p \quad (n \geq 4),$$

and consequently $e_0(T_p) = 24p$ and $e_1(T_p) = 39p$.

Proof. Theorem 13.1 supplies the square. The residue identity $\text{low}(A_p + C_p) = [0, s - 2]$ then gives

$$v(T_p^3) = [12s, 13s - 2] \cup [13s, \infty).$$

Adding $4s + A_p$ fills $[16s, 17s - 2]$ and the tail from $17s$; because $1 \in A_p$, $(4s + 1) + (13s - 2) = 17s - 1$ fills the intervening endpoint. Hence

$$v(T_p^4) = [16s, \infty), \quad v(T_p^5) = [20s, \infty) = v(Q_p T_p^4).$$

Multiplication proves the same equality for every later power. Subtracting the shifted square and cubic profiles gives

$$\begin{aligned} v(T_p^3) \setminus v(Q_p T_p^2) &= 12s + ([2p + 1, 3p - 1] \cup [5p - 1, 6p - 2]) \cup \{17s - 1\}, \\ v(T_p^4) \setminus v(Q_p T_p^3) &= \{17s - 1\}. \end{aligned}$$

Their cardinalities are $2p$ and 1 . Theorem 13.1 gives $14p$ at the preceding stage. Finally, subtracting $v(Q_p) = 4s + \Gamma_p$ from Theorem 12.1 leaves eight disjoint blocks of sizes

$$2p - 1, \quad 5p, \quad 3p, \quad 2p, \quad 3p, \quad 6p, \quad 2p - 1, \quad 1,$$

whose sum is $23p - 1$. The strict nonzero quotients and terminal equality prove reduction number four. Since T_p is maximal-ideal primary in dimension one, the one-generated reduction is minimal.

The parameter ideal Q_p has colength $4s = 24p$. Multiplication by its nonzerodivisor generator gives

$$\ell(T_p^i / T_p^{i+1}) = 24p - \ell(T_p^{i+1} / Q_p T_p^i).$$

Summing through $i = n - 1$ and using $(23p - 1) + 14p + 2p + 1 = 39p$ proves the Hilbert formula and coefficients. $\square$

Two exact representations confirm every displayed set and length for all $p = 4, \dots, 300$ [MV]. A separate tail-set implementation rehashes all 297 rows, reconstructs six parameters, and rejects deleted-terminal, false-stabilization, altered-interval, and perturbed-length controls. The campaign and audit aggregates are

e9c3c887648f08cf67c614b381f00c8c6520dcd1bb89f8cdece62293bfd06030

and

0f6ed70676fffb8972b8b167ad52c4f9d2851f69c3b1d96f4023e5e3d5825c781.

15 Reproducibility and trust boundary

All compact code, hypotheses, manifests, audits, and verdicts are in the public repository under

[problems/commutative-algebra/huneke-wiegand/](https://github.com/caos-problems/commutative-algebra/huneke-wiegand/).

Heavy CNFs and proofs are retained outside Git with hashes committed in the manifests. The minimum-layer archive adds 12 fully accounted files and a reconstruction auditor. The record also includes the compact EXP-011–017 campaigns, independent reconstruction audits, symbolic proofs, the preserved EXP-013 correction, and the EXP-015 refutation. The proof toolchain is pinned as follows:

Component	Identity
CaDiCaL	reported 1.7.3; Ubuntu package 1.7.4-1
CaDiCaL binary SHA-256	7b73df0a6d9cf3c751a1948300e5baff8e82c4d39bcd88f0c063b5f5cfb8b33e
DRAT-trim commit	2e3b2dc0ecf938addbd779d42877b6ed69d9a985
DRAT-trim binary SHA-256	92f0aa9575ed519d66a99b8b1b3dde6ece4618ae4c202a3a4b200265dda0aa7a

The trusted base comprises the cited tree theorem, the finite reduction proved here, the small standard-library model checker, and DRAT-trim’s verification of each proof against its exact CNF. The SAT solver need not be trusted for UNSAT. For SAT, the independently extracted model is checked without consulting solver internals.

16 Scope, attribution, and open questions

Son Pham retains discovery priority for the counterexample. Professor Craig Huneke’s verification is independent external evidence. The contributions proved here are Frobenius minimality, uniqueness of the normalized pair at that minimum, the separate infinite family in Theorem 9.2, and its uniform endomorphism anatomy in Theorem 10.1, pseudo-Frobenius anatomy in Theorem 11.1, exact trace/conductor formula in Theorem 12.1, stability defect in Theorem 13.1, and conductor reduction/Hilbert data in Theorem 14.1, together with their reproducible records.

Theorems 6.1, 7.2, 9.2, 10.1, 11.1, 12.1, 13.1, and 14.1 do not classify all higher Frobenius values, prove a minimum multiplicity or embedding dimension across all counterexamples, or establish corresponding statements for arbitrary Gorenstein domains and modules. The following questions remain open:

1. How does the number of normalized rigid pairs grow above $F = 181$?
2. Is there a global minimum under multiplicity, embedding dimension, or number of minimal generators?
3. Which other Boolean-scaffold models lie on affine or residue-class families, and is the family in Theorem 9.2 maximal inside a natural Kunz face?
4. Which additional hypotheses on the endomorphism ring recover a positive theorem while excluding both the public example and the full family proved here?
5. Can the finite certificates be imported into a smaller formally verified checker?

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